

Adversarial Review

Structured disagreement for grounded code review

Agenda

- 01** The design question
- 02** Building AR step by step
- 03** Where AR breaks
- 04** The one-prompt fix
- 05** Real repositories and cost

| How many agents?



Adversarial Review

A reviewer writes. A critic audits the review.

Three benchmarks, three wins

87%

LiveCodeBench

0.533

SWE-PRBench

75.2%

SWE-bench

3

Agents

Three contributions

A minimal reviewer-critic loop works

Agents optimise for agreement

Structured disagreement fixes it



01

The design question

Two paradigms

Role-separated teams

Many agents, split by role

Diminishing returns

Coordination overhead

Subagents as tools

Invoked like a tool call

No agent interaction

The production default

| Is there a middle?

Prior work, in three families

Self-refinement

One model critiques itself

Role-separated teams

MapCoder, AgentCoder, CodeCoR

Debate and aggregation

MAD, then MARS

Debate versus aggregation

MAD

Everyone reads everyone

Cost grows with agents

Pass-rate can drop

MARS

Reviewers run in parallel

A meta-reviewer decides

No interaction at all

Three benchmarks, three claims

	LCB	PRBench	Verified
Task	Generate	Review	Repair
Metric	Pass %	F1	pass@1
Size	105	100	500



02

Building AR

The LiveCodeBench setup

105 stdin-style tasks

57 tagged hard

Claude Sonnet 4.5, medium

One design choice per step

Where the methods separate



The construction ladder

<u>01</u>	<u>02</u>	<u>03</u>	<u>04</u>
Zero-shot	Refine	Reviewer	Two more
No check	Self check	One outside	In parallel

77%

The starting pass-rate

Four methods land on this number.

Why self-critique fails

Critic and generator are one model

It repeats its own mistakes

So it misses what it missed

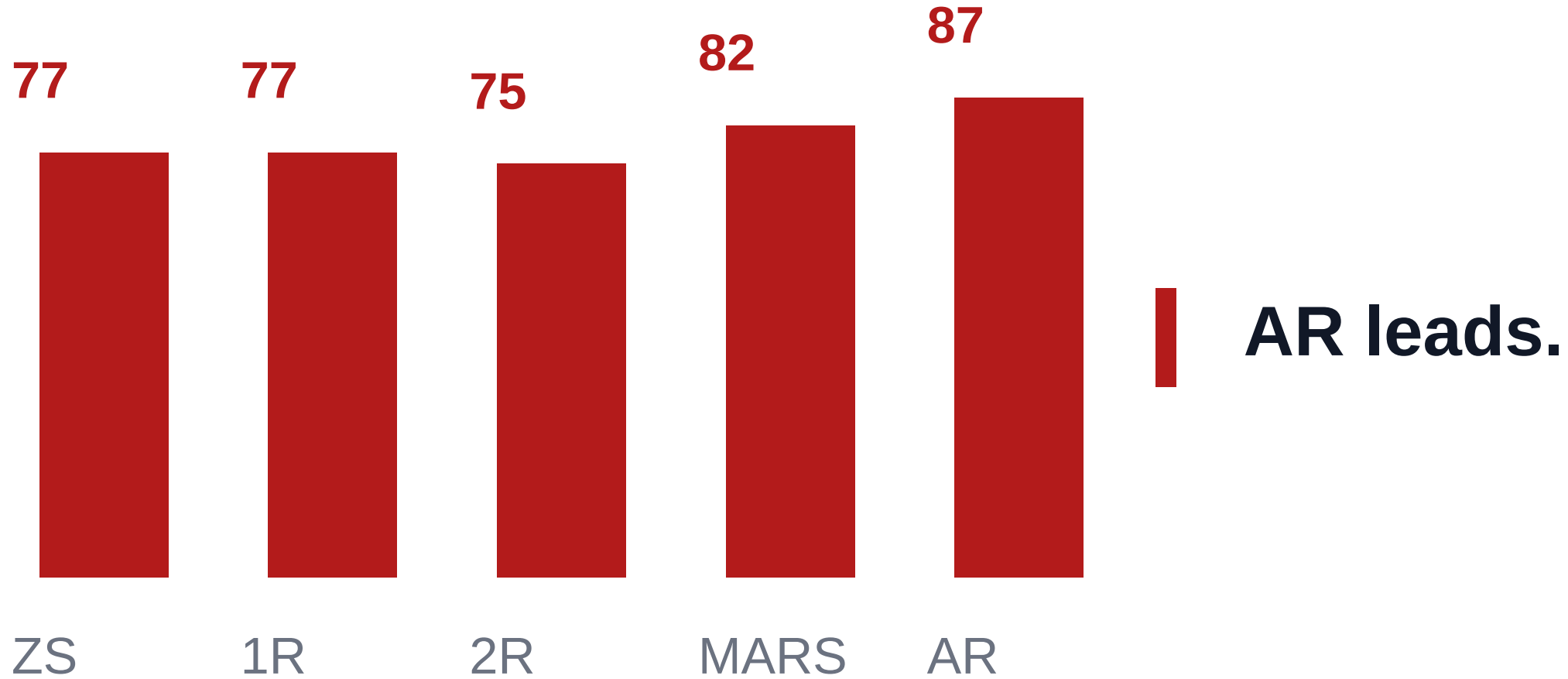
Why one reviewer is not enough

High variance across runs

It can approve without checking

Or flag nits and miss the bug

The cluster, and the break



MARS spends five agents

Author

Writes the
draft

Reviewers

Three in
parallel

Meta

One final
verdict

Cap

Two rounds
only

MARS breaks the cluster

82%

Pass-rate

+5 pts

39/57

Hard tasks

+3

5

Agents

+4

But MARS pays for it

Five agents for a small gain

All interaction is removed

Parallel reviews never meet

| Can two agents talk?



03

Inside the loop



The inner loop

Review text moves. The artifact never does.

The inner loop

01 Reviewer

R inspects the frozen artifact

02 Critic

A fresh C audits that review

03 Repeat

Until R and C are consistent

The outer loop

01 Converge

One consistent review emerges

02 Decide

Accept it, or hand it to M

03 Edit

M writes artifact version $N+1$

First-pass termination

Reviewer approved at once

Critic agreed at once

No back-and-forth needed

Otherwise, edit and repeat



"Self-review is not review."

- AR SKILL.md

Three roles, one interaction

M

Writes, edits

R

Reviews code

C

Audits R

LiveCodeBench results

Method	Pass	Agents
Two-reviewers	75%	3
Zero-shot	77%	1
MARS	82%	5
AR	87%	3

87%

AR on LiveCodeBench

Five points over MARS.

Reviewing roles

4

MARS

VS

2

AR



04

Review quality

The SWE-PRBench setup

100 real pull-request diffs

A GPT-5.2 judge scores them

Kappa 0.75 against humans

F1 over matched comments

SWE-PRBench F1

Method	F1	N
AR	0.457	100
Single-reviewer	0.495	100
MARS	0.501	100
Two-reviewers	0.503	100

0.457

AR, naively applied

Last of the four review methods.

| Why did AR lose?

Two failure modes

Over-decomposition

The reviewer hedges

The critic confirms them

Thin comments pile up

The critic yields

C finds a real bug

R rebuts, cites no code

C gives in anyway

Case A: what survives

Flags raised · 5

Judged real · 2

Actual bugs · 1

Case B: the dropped bug

0.286

AR on this task

0.667

MARS on this task



False consensus

Agreement without the evidence to earn it.

A reliability bug, not a score

It looks like independent checks

It comes from the interaction

Not from weak agents



"Agreement is not validation."

- Section 7



05

The one-prompt fix

Three verdicts, not two

01 AGREE

Every flag is real

02 EVIDENCE

Cite code that contradicts

03 CONCERN

Doubt with no code to cite



The critic's output format

Output format

AGREE

DISAGREE_EVIDENCE: <cite file:line>

DISAGREE_CONCERN: <plausible, unproven>

What the reviewer owes back

AGREE

Keep every flag

EVIDENCE

Revise the flag

CONCERN

Cite code or drop

The text constraint pays

0.533

AR constrained

+0.076

0.503

Best baseline

+0.030

What did not change

Still one reviewer, one critic

Same wiring, same agent count

Only the prompt text changed



06

Real repositories

The SWE-bench setup

500 real GitHub issues

Claude Code, full tool access

AR written as a SKILL.md

pass@1 via the Docker harness

SWE-bench Verified

Method	Pass@1	N
Zero-shot	71.6%	500
MARS	72.6%	500
AR	75.2%	500

75.2%



AR on real issues

No orchestrator, just the text protocol.

Case 1: one frame deeper

Round 1

R blames caller

Symptom

Round 1

C pushes back

Root cause

Edit

M fixes callee

20 lines

Case 1: symptom or cause

Zero-shot and MARS

AR

Patch the calling code

Patch the callee instead

45 and 50 line patches

About 20 lines

Both fail hidden tests

The only one that passes

Case 2: scope creep

astropy, add one argument

Zero-shot ships 24 lines

AR ships 32 and over-edits

Nobody called it out of scope

When AR hurts

24

Zero-shot lines

VS

32

AR lines



07

Cost and limits

4.5x

AR's token multiple

Measured against the zero-shot scaffold.



Pareto frontier

Nothing tested is cheaper and better.

When to spend the tokens

Complex repositories

Root cause beats token cost

Skip it on easy tasks

AR in one frame

Strengths

Top score on all three

Opportunities

Scope-aware critics next

Weaknesses

About 4.5x the tokens

Threats

Judges reward surface

What the paper does not claim

No formal guarantee

Three probes, not full coverage

Each benchmark sees one angle

Future work

A scope-checking critic

A gate that skips easy tasks

Training on disagreement traces

The takeaway

Do not scale the agent count

Make disagreement explicit

Ground every flag in code



"Minimal, structured, grounded."

- Qiu and Gill, 2026